Contractual Governance of Public Works: Evidence Production and Monitoring in the UK and Japan

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26th International Conference of Group Decision and Negotiation

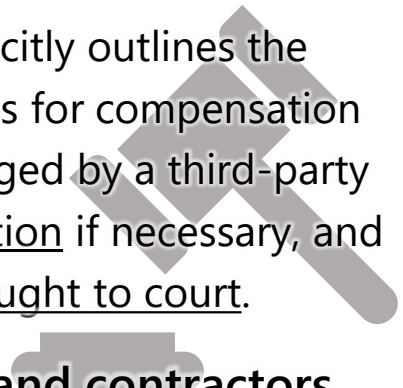
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Background:
Institutional Contrast in Contract Change of Public Construction Works

The UK:
Formal claims and frequent disputes

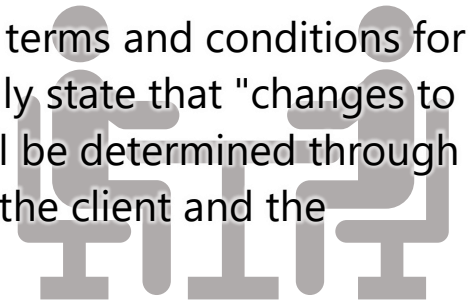
1. The triggers and procedures for contract changes are **explicitly defined.**
2. **A third party** makes a decision on contract changes
   - NEC ECC clause 60.1 explicitly outlines the conditions and procedures for compensation events. These are first judged by a third-party PM, followed by adjudication if necessary, and then the case may be brought to court.



- **Issue**: Traditionally, **clients and contractors have been prone to conflict**, leading to project delays and increased transaction costs due to disputes.

Japan:
Consultation and client-side adjustment

1. Contract changes are **processed in consultation between the client and the contractor**.
2. **The government client** makes a decision of contract changes
   - The standard contract terms and conditions for public construction only state that "changes to the contract price shall be determined through consultation between the client and the contractor."



- **Issue**: Traditionally, government staff have possessed high technical capabilities. Recently, **a decline in the capabilities of client organizations** raises concerns that the traditional system may become unsustainable.

Research Question and Approach

The systems and challenges around contract governance in the UK and Japan are contrasting.

Can we solve the problem through a combination of both systems? Or is that difficult?

Research Approach: a qualitative interpretation based on analytical game-theoretic models

- We examined the conditions and characteristics of UK/JPN systems with equilibrium analysis.
  1. Contract Game @Transactional Domain: with Extensive-Form Games
  2. Skill-Investment Game @Organizational Field: with Strategic Games

  } 3. Comparative Institutional Analysis
  We connect the two games through institutional linkage

▶ ▶Policy insights that will contribute to resolving the institutional challenges

Related Literature and Contribution

Basic Ideas & Previous studies

- Comparative Institutional Analysis: Aoki (2001) "Toward a Comparative Institutional Analysis"
- Incentive Game:
  - Costly Litigation: Choi et al. (2004) "Completing Contracts in the Shadow of Costly Verification"
  - Relational Contract: Levin (2003) "Relational Incentive Contracts"
  - Subjective Assessment: Baker et al. (1994) "Subjective Performance Measures in Optimal Incentive Contracts"

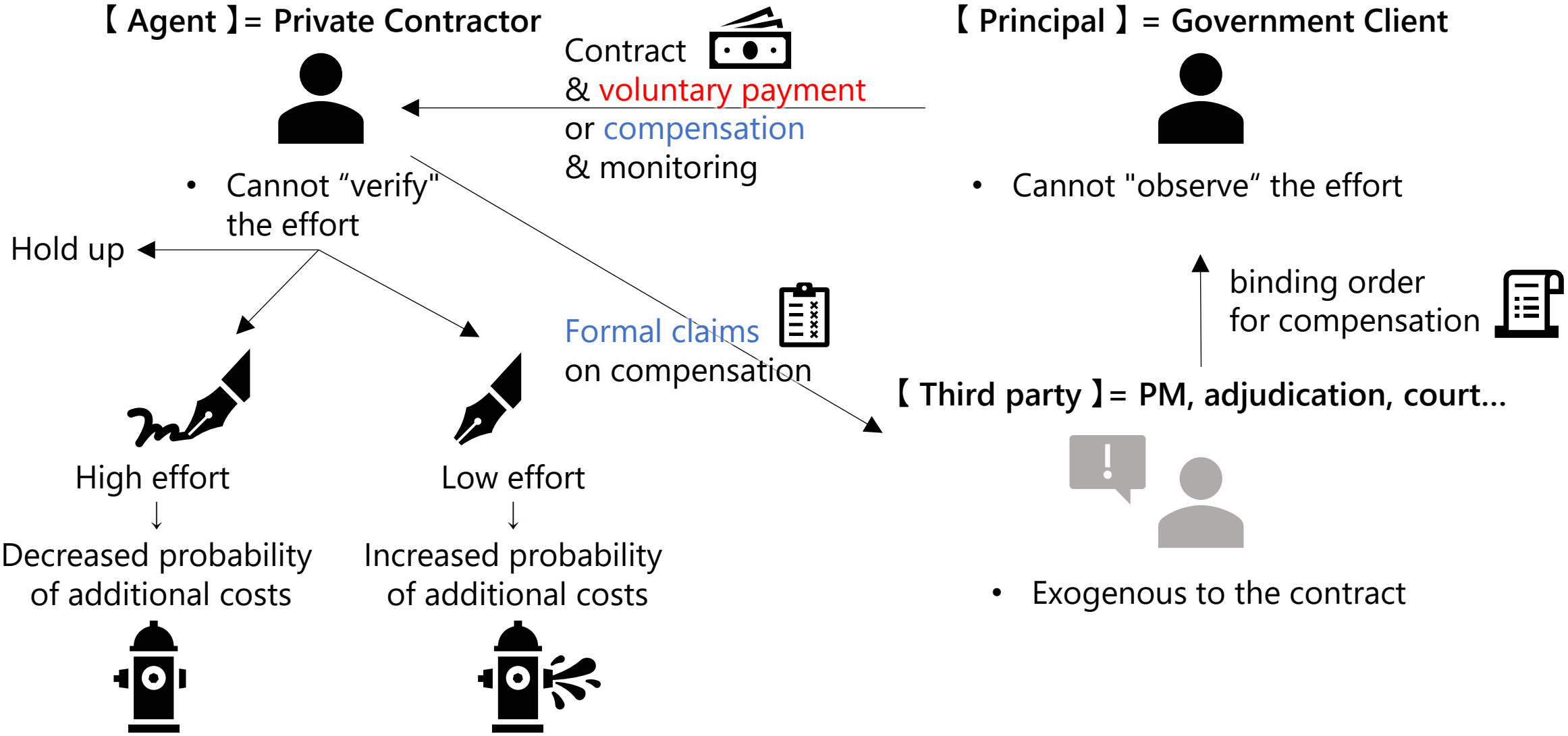
Novelty of this research

- Analyzing social issues in public procurement contracts in the UK and Japan from a game-theoretic perspective.
- **Examining the complementarities between contract governance and skill-investment through institutional linkages.**
- Applying concepts of "Subjective Assessment", "Relational Contract", and "Costly Litigation", which are all from incentive games, to public construction works contracts.

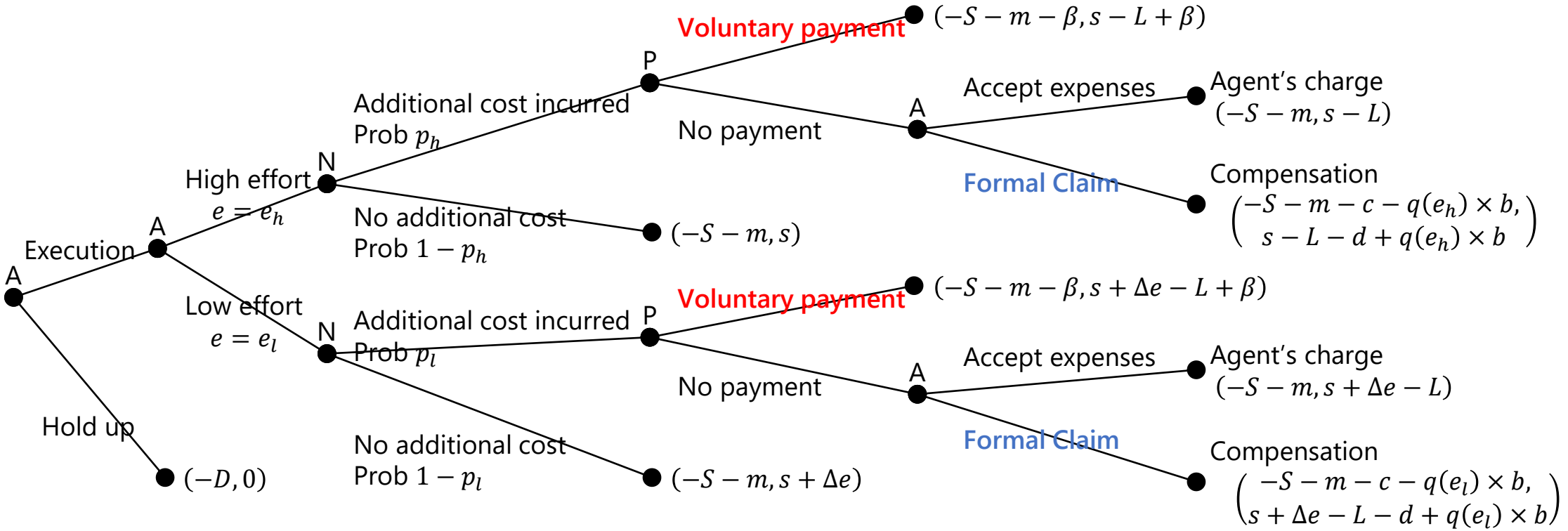
1) Contract Game @Transactional Domain

- Concept
- UK-type
- Japanese-type
- Main Theoretical Results

Contract Game: the Players and the Story



Contract Game: Extensive-form Game



Legends
●: Decision-making node
N : Nature (stochastic transition)
P : Principal (government client)
A : Agent (contractor)
(Principal's payoff, Agent's payoff)

S: Initial contract amount
L: Additional costs necessary to execute the project
e = {e_l , e_h} : Agent effort cost
Δe = e_h - e_l
s=S-e_h: Agent's margin when high effort
p = {p_l, p_h}: Probability of additional costs
d: Agent's evidentiary costs
c: Principal's claim response cost

q(e) : Probability that the third party approves formal claim
b: Compensation amount awarded through formal claim
β: Principal's voluntary additional payment amount
D: Principal's loss when the project cannot be carried out

One-shot Game: Dividing into Cases for Subgame Equilibria

Agent Claim or not \ **Principal** Pay or not	**X.1 Claim is cheaper:** $-\beta < -c - q(e_h) \times b$ -> **Never pay**	**X.2 Depends…:** $-c - q(e_h) \times b \leq -\beta < -c - q(e_l) \times b$ -> **If high effort, pay; if low effot, do not**	**X.3 Cheaper to pay:** $-\beta \geq -c - q(e_l) \times b$ -> **Always pay voluntarily**
1. Claims never pay: $-d + q(e_h) \times b < 0$ -> **Never claim**	1.1 No claim	N/A	N/A
2. Claims pay, if effort: $-d + q(e_h) \times b \geq 0 > -d + q(e_l) \times b$ -> **If high effort, claim; if low effort, accept.**	2.1 High effort + claim on compensation	No subgame equilibrium	N/A
3. Claims are profitable: $-d + q(e_l) \times b > 0$ -> **Always claim.**	3.1 High or low effort + claim on compensation	No subgame equilibrium	3.3 High or low effort + voluntary payment by the principal

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So where are the UK and Japan?

In Japan, "agents will never claim" + "principals may voluntarily pay" ... N/A!!
-> Introduction of "Relational Contract"

Agent / Principal (Claim or not / Pay or not)	X.1 Claim is cheaper: $-\beta < -c - q(e_h) \times b$ -> Never pay	X.2 Depends...: $-c - q(e_h) \times b \leq -\beta < -c - q(e_l) \times b$ -> If high effort, pay; if low effot, stay	X.3 Cheaper to pay: $-\beta \geq -c - q(e_l) \times b$ -> Always pay voluntarily
1. Claims never pay: $-d + q(e_h) \times b < 0$ **-> Never claim**	1.1 No claim	N/A	N/A
2. Claims pay, if effort: $-d + q(e_h) \times b \geq 0 > -d + q(e_l) \times b$ **-> If high effort, claim; if low effort, accept.**	2.1 High effort +claim	No subgame equilibrium	N/A
3. Claims are profitable: $-d + q(e_l) \times b > 0$ **-> Always claim.**	3.1 High or low effort + claim	No subgame equilibrium	3.3 High or low effort + voluntary payment by the

In the UK, "agents can claim" + "principals will not pay," so here

UK Equilibria: Formal Claims and Evidence Production

In the UK, the principal never makes voluntary payments, and the agent may claim.

- Expected compensation relaxes the agent's participation constraint.

Equilibrium 2.1 & 3.1

Agent
- **High effort** + claim

Principal
- No voluntary payment

Improvement of evidentiary efficiency:
$-d + q(e) \times b$

d: Agent's evidentiary costs
q(e): Approval rate of compensation
b: Compensation amount

Equilibrium 3.1

Agent
- **Low effort** + claim

Principal
- No voluntary payment

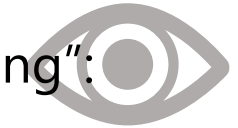
→ Given the shape of $q()$, we can specify an equilibrium.

→ **If d and $q()$ change over the long term, the agent will improve its evidentiary efficiency. Eventually he may reach a low-effort equilibrium with improved $q()$.**

Japanese Equilibria: Relational Payment and Monitoring

- Let's introduce **"Relational Contract" (a self-enforcing repeated game)** such as:

> - The agent will choose high effort.
> - If additional costs are incurred, the principal will voluntarily pay β under certain conditions.
> - If no additional costs are incurred, or if additional costs are incurred but paid, the relational contract will continue.
> - If additional costs are incurred but not paid, the relational contract will be terminated, and the game will continue as one-shot games.



- As "certain conditions" for this principal's voluntary payment, we introduce "monitoring":
  - The principal pays a cost m to observe the stochastic signal of the agent's efforts. $z = 1$ is a subjective signal indicating the agent's effort, and $z = 0$ is a negative signal.
    The principal promises to pay if she observes $z = 1$.
  - Accuracy of monitoring: $a_h \coloneqq \mathrm{Prob}(z = 1 | e = e_h) > a_l \coloneqq \mathrm{Prob}(z = 1 | e = e_l)$

→Result of equilibrium alaysis:

With the above relational contracts, Japanese equilibria can be established.

Japanese Equilibria – Result of the Equilibrium Analysis

Principal

- Expected payoff from relational contract $U^* = {}^{1}/_{1-\delta(1-p_h)}\{-S - m + \delta p_h \bar{U}\}$.
  Therefore, a smaller monitoring cost m increases the principal's payoff.
  - In particular when $m \geq {}^{\delta\Delta p(D-S)}/_{1-\delta(1-p_l)}$, a non-monitoring equilibrium with low effort is better than monitoring.
- IR (Participation constraint) & IC (Incentive compatibility) $\beta = {}^{\delta}/_{1-\delta(1-p_h)}\{-S - m + D\} \geq 0$

Agent

- Required: IR constraint $\beta \geq L - {}^{s}/_{p_h}$
- **IC constraints <u>without monitoring</u>**: 
  $\min\left\{{}^{\delta}/_{(1-\delta)}\left({}^{\Delta e}/_{\Delta p}\, p_h - s\right), L - {}^{\Delta e}/_{\Delta p}\right\} \geq \beta,$ or $s - {}^{\Delta e}/_{\Delta p}\, p_h \geq 0$
- **IC constraint <u>with monitoring</u>**: 

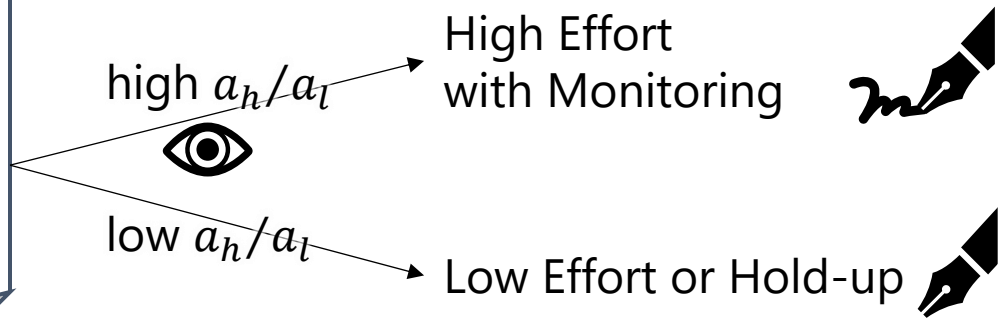
Can be relaxed

$$[\Delta p L + \beta(p_h a_h - p_l a_l) - \Delta e][1 - \delta + \delta p_h(1 - a_h)] + \delta[\Delta p + (p_h a_h - p_l a_l)][s + p_h(-L + a_h \beta)] \geq 0$$

Japanese Equilibria – Result of the Equilibrium Analysis

- Key Result:
  1. Even when random payment cannot satisfy the agent's IC condition, **monitoring can restore high-effort incentives** if a_h/a_l, which represents the principal's monitoring capability to distinguish low-effort from high-effort, is sufficiently large.
  2. However, if monitoring accuracy (a_h/a_l) is low, difficult projects (with large L, large D, small S, large $\Delta e/\Delta p$, and large p_h) may remain low effort equilibrium or hold up.

"Difficult projects"
- large L (stochastic additional cost)
- large D (project value)
- small S (original contract price) = Difficult Project
- large $\Delta e/\Delta p$ (risk reduction cost)
- large p_h (loss probability with high effort)



- **Client-side monitoring can make these difficult projects feasible/executable.**

Main Theoretical Results from Contract Game

What we did:

- **Developed a unified game tree** which can accommodate both UK case and JPN case
- Showed that UK and JPN systems are **variations of equilibria belonging to one game tree**

- Proposition 1 from the UK equilibria: 
  A formal claim route **relaxes the agent's participation constraint, but it may weaken effort incentives** if compensation becomes too easy to obtain.

- Proposition 2 from the Japanese equilibria: 
  Relational contracts with voluntary payment by the principal can **make difficult projects executable**, but it **requires sufficient client-side monitoring** capability.

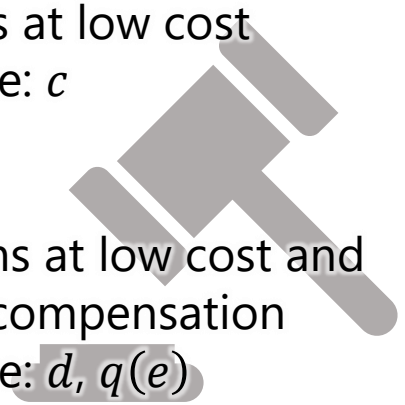
2) Skill-Investment Game @Organizational Field

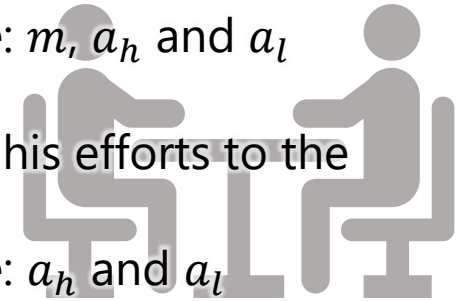
- Concept
- UK-type
- Japanese-type

What Do We Mean by “Skills"?

- In this Skill-Investment Game, players can invest to develop their capabilities, which can improve their parameters/functions in the Contract Game.

	Formal-claims skills: The ability to process formal claims regarding compensation for additional costs in accordance with the rules.	Relational coordination skills: The ability to send and receive information about agent efforts, which is private information in contract game.
For the principal…	• to handle claims at low cost • In contract game: c	• to monitor and identify the efforts of the contractor • In contract game: m, a_h and a_l
For the agent…	• to prepare claims at low cost and efficiently gain compensation • In contract game: d, $q(e)$	• to communicate his efforts to the client • In contract game: a_h and a_l





Concept of Skill-Investment Game Model

- Let's assume a strategic game between the principal and the agent
  - $\boldsymbol{u_{aa}^{i}}$= -Skill investment costs + Gain from Contract game

Principal \ Agent	Investing in formal-claims skills	Investing in coordination skills	No investment in skills
Investing in formal-claims skills	(u_{ss}^{P}, u_{ss}^{A})	(u_{sc}^{P}, u_{sc}^{A})	(u_{SN}^{P}, u_{SN}^{A})
Investing in coordination skills	(u_{cs}^{P}, u_{cs}^{A})	(u_{cc}^{P}, u_{cc}^{A})	(u_{cN}^{P}, u_{cN}^{A})
No investment in skills	(u_{NS}^{P}, u_{NS}^{A})	(u_{Nc}^{P}, u_{Nc}^{A})	(u_{NN}^{P}, u_{NN}^{A})

Purpose of this model

1. Reproduce the current skill strategies in the UK and Japan as equilibria
2. And see what happens to the equilibrium strategies, when conditions change
   - UK: transition to low-effort equilibrium in contract game
   - JPN: The principal stops investing in skills

UK: Investment in Formal-Claims Skills

- Skill-Investment game in the UK

Principal \ Agent	Formal-claims skills	Coordination skills	No invest
Formal-claims skills	(1 , 1)	(0, -1)	(0, 0)
Coordination skills	(0, 1)	(-1, -1)	(-1, 0)
No invest	(1 , 1)	(0, -1)	(0, 0)

Low effort equilibrium ▶▶

Principal \ Agent	Formal-claims skills	Coordination skills	No invest
Formal-claims skills	(-1, 1)	(0, -1)	(0, 0)
Coordination skills	(-2, 1)	(-1, -1)	(-1, 0)
No invest	(-1, 1)	(0, -1)	(0, 0)

Though the principal's gain decreases,
She cannot escape from the equilibrium strategy.

- Setting the cost & payoff

	Investment in…	Principal's cost	Principal's gain	Agent's cost	Agent's gain
Principal's	Formal-claims Skill	-1	+1 Reduced claim process costs	0	0
	Coordination Skill	-1	0	0	0
Agent's	Formal-claims Skill	0	+1 ▶▶ -1 Avoiding hold up by improved IR ▶▶ shift to low-effort equilibrium	-1	+2 Improved IR leads to avoidance of hold up and improved gains.
	Coordination Skill	0	0	-1	0

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JPN: Investment in Relational Coordination Skills

- Skill-Investment game in Japan

Principal \ Agent	Formal-claims skills	Coordination skills	No invest
Formal-claims skills	(-1, -1)	(0, -1)	(-1, 0)
Coordination skills	(0, 1)	(2, 2)	(1, 1)
No invest	(0, -1)	(1, 1)	(0, 0)

Principal quits to invest ▶▶

Principal \ Agent	Formal-claims skills	Coordination skills	No invest
Formal-claims skills	-	-	-
Coorrdination skills	-	-	-
No invest	(0, -1)	(1, 1)	(0, 0)

The agent may remain locked into relational coordination capability

- Setting the cost & payoff

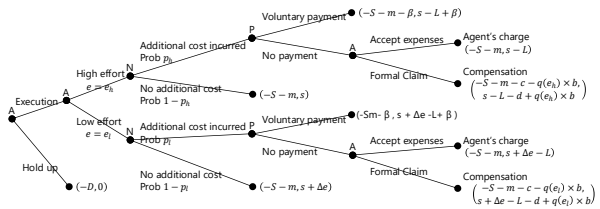
	Investment in…	Principal's cost	Principal's gain	Agent's cost	Agent's gain
Principal's	Formal-claims Skill	-1	0	0	0
	Coordination Skill	-1 ▶▶ 0 ▶▶ Stop investment	+2 ▶▶ 0 avoidance of low-effort equilibrium and hold up through improved agent IC ▶▶ No investment	0	+1 ▶▶ 0 IC improvements lead to avoidance of hold up and increase payoff ▶▶ Principal no longer invests
Agent's	Formal-claims Skill	0	0	-1	0
	Coordination Skill	0	+1 avoidance of low-effort equilibrium and hold up through improved agent IC	-1	+2 IC improvements lead to avoidance of hold up and increase payoff

3) Institutional Linkage

Institutional Linkage between Contract Governance and Skill Investment

- Capability levels of the players

① Contract Game
Short-term @Transactional Domain



② Skill-Investment Game
Long-term @ Organizational Field

Principal \ Agent	Formal-claims skills	Coordination skills
Formal-claims skills	(-1, -1)	(0, -1)
Coordination skills	(0, 1)	(2, 2)

- Gains from contractual game

UK	Active formal claims	**COMPLEMENTARY**	Formal-claims skills
JPN	Voluntary payment		Relational coordination skills

- Proposition 3: Given equilibria in the contractual game, the different strategies of the UK and JPN in the Skill-Investment game are rational, and vice versa.

Policy Implications

1. About the Current Social Challenges



- UK: **If contractors improve their evidence-production capability too much**, there is a risk that
  1) contractors will **frequently seek formal claims**, which increases transaction costs,
  2) **leading to a low-effort equilibrium**



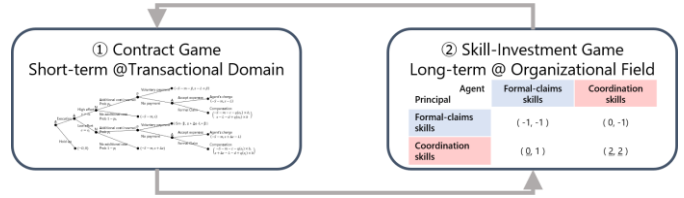
- Japan: **A decline in the government client's investment in skills** (monitoring capability in the contract game) could **make it impossible to undertake large-scale and complex projects**.

2. About Potential Solutions

- The current system relies on complementary relationships between games across multiple domains, **making a transition to a different system—i.e., a different equilibrium—difficult.**
- For a transition to a different equilibrium to occur, skill accumulation needs to exceed a threshold, so players must continue investing in the alternative capability even when it is not immediately profitable.

➡ Institutional reform cannot be limited to contract clauses alone.
It also requires capability formation on both the client and the contractor.

Conclusion and Limitations



- What we found in this research:
  - **Contract governance and skill investment are mutually linked**
    - **Different contractual systems generate different incentives for capability formation, and these capabilities, in turn, stabilize the original contractual systems.**
- Limitations
  - Overall Model Structure
    - Assumption of risk-neutral players
    - No comparison of total surplus across equilibria
  - Contractual Game Model
    - Simplification of “consultation between the client and the contractor”
    - A stable relational contract with fixed terms may be unrealistic
    - Random compensation without monitoring is not realistic in terms of government accountability
  - Skill-Investment Model
    - Oversimplification of payoffs